BBO capstone project presentation template

Develop a concise presentation that captures:

1. **An overview of your BBO approach**
   - Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
   - Briefly describe your overall strategy and how your process fits together.

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OBJECTIVE
I aim to maximise eight unknown functions with a limited evaluation budget: one new input per function per round. Inputs have two to eight coordinates. I observe the returned numerical result but cannot inspect the underlying function.

CURRENT EVIDENCE
My dataset contains 175 supplied observations plus 88 evaluations from eleven completed rounds: 263 evaluated input-output pairs. Function-specific datasets contain 21 to 51 observations. Eight round-12 queries are prepared and validated; their outcomes are not yet available.

HOW MY PROCESS FITS TOGETHER
I validate the cumulative data, compare the latest outputs with the previous best, and review historical prediction errors. I then fit a separate Gaussian Process (GP) to each function using a Matern kernel and target normalisation.

The GP estimates a predicted result and uncertainty. I rank candidate inputs using UCB = predicted mean + 2 x predictive standard deviation. This balances promising predictions with opportunities to learn in uncertain regions.

Currently, I compare 100,000 global candidates with 10,000 candidates in each of two local boxes around the best observation. I validate the selected coordinates, retain their predictions, and prepare six-decimal strings for manual portal submission.

Only returned evaluations count as results. A high UCB is a reason to test a point, not a guarantee of improvement.

2. **How your strategy has evolved**
   - Describe the key changes in your approach since your early rounds.
   - What influenced these changes – data trends, feedback, model performance or intuition?
   - Identify which principles or heuristics now guide your query decisions.

FROM A BASELINE TO A DOCUMENTED WORKFLOW

In CP_12, I used GP-UCB with global random candidate search. Subsequent notebooks compared original and wider kernel bounds, responding to fitted parameters reaching their limits. A boundary warning motivated investigation rather than proving a model was wrong.

CP_14 added linear and RBF SVM diagnostics. CP_15-CP_17 compared classification and regression approaches, including linear models and small neural networks. These experiments helped distinguish predictive checks from the continuous optimisation objective. The current auxiliary models are an eight-unit neural network for F5 and linear regression for F7; they do not supply UCB uncertainty.

CP_18 compared Matern smoothness and bound configurations using cross-validation RMSE. Later notebooks retained the selected configurations while re-estimating continuous kernel parameters as feedback arrived. Global and local proposals made the exploration-versus-refinement choice explicit.

CP_19 compared simple and structured prompts around supplied candidate data. Numerical candidate ranking remained in Python. CP_21 strengthened transparency and documentation; CP_22 added regional distance diagnostics and an F5 minimum-distance constraint. CP_23 added PCA and reproducible candidate seeds that vary by function and round.

WHAT NOW GUIDES MY DECISIONS

I use observed progress, prediction errors, candidate distances and model disagreement. I preserve negative and near-zero outputs, avoid double-counting cumulative feedback, and separate completed experiments from planned changes. No function improved in round 11, reinforcing the need to examine redundancy and uncertainty rather than equating more data with automatic gains.

3. **Patterns, data and insights**
   - Highlight the most meaningful trends you've observed across your data points or model evaluations.
   - Identify which variables or behaviours seem to influence results the most.
   - Explain how these observations have shaped your understanding of the search process.

OBSERVED PROGRESS THROUGH ROUND 11

Seven functions improved over their initial best. Values below are rounded and must be compared within each function:

Function	Initial best	Current best
F1	7.71e-16	7.71e-16
F2	0.611205	0.658240
F3	-0.034835	-0.030964
F4	-4.025542	0.615273
F5	1088.86	8515.38
F6	-0.714265	-0.242431
F7	1.364968	2.436323
F8	9.598482	9.958227

Progress is uneven: five functions improved in round 1, two in round 10, and none in round 11. These counts do not measure the size of gains or establish convergence.

PATTERNS AND THEIR LIMITS

F5's strongest observations are near the upper input boundaries. F4 has a repeatedly sampled promising interior region; F6's best point has its fifth coordinate close to zero. These associations reflect adaptive sampling and do not establish causal variable importance.

PCA describes input variation, not output sensitivity. The first two components retain 55.1% of input variance for F7 and 47.0% for F8. Five and seven components are needed for 90%, respectively, so I retained all coordinates.

For F3, x3 has a small PC1 weight but a short GP length scale. This contrast shows why a low-variance input direction can still matter to the objective. F1's extremely small outputs also mean rank colours can exaggerate practically tiny differences.

4. **Decision-making and iteration**
   - Explain how you balance exploration and exploitation.
   - Describe one or two examples of strategic decisions. What worked, what didn't, and why?
   - Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

BALANCING EXPLORATION AND EXPLOITATION

UCB rewards both predicted performance and uncertainty. For round 12, the highest eligible scores select global proposals for F1, F3 and F7, and local proposals for F2, F4, F5, F6 and F8. Search labels describe candidate generation, not proven global or local optima.

EXAMPLE 1: REDUCING REPETITION IN F5

A round-10 query only about 0.000235 from an earlier observation improved the best output by 5.089, to 8515.379. The gain was real but small. In round 11, I required a Euclidean separation of at least 0.01; the selected point returned 8311.706 and did not improve the best.

The constraint therefore has no demonstrated performance advantage. The unsubmitted alternative was not evaluated, so its result is unknown. I retained the constraint for round 12 to reduce repetition: separation is 0.010475, at a UCB reduction of 0.715% versus the unconstrained winner.

EXAMPLE 2: ACCURACY IS NOT IMPROVEMENT

For F7 in round 11, the GP predicted 2.388085 and observed 2.351393: absolute error about 0.0367. Linear regression predicted 1.206557, an error about 1.1448. The GP was more accurate at this point, but the output still fell below the incumbent 2.436323.

HANDLING SURPRISES

I retain unsuccessful evaluations, refit the models and record bound warnings. Error divided by predicted standard deviation is a diagnostic, not a full calibration test. White-noise contributions and model mismatch mean predictive uncertainty is not entirely reducible through more sampling.

5. **Next steps and reflection**
   - Outline your planned actions or experiments to improve the model's performance.
   - Include two or three sentences connecting your work to the broader ML landscape.
   - Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

NEXT: THE FINAL ROUND IN MODULE 24
When round-12 feedback arrives, I will match it to the submitted coordinates, compare predictions with observations and update the best-so-far history. I will examine whether distant proposals discovered useful regions and whether nearby proposals added meaningful gains.

For the final submission, I will consider local refinement around newly successful points, while preserving exploration where predictions remain weak. I will reassess F5's distance constraint using the accumulated evidence rather than assuming that greater separation is always beneficial.

PLANNED EXPERIMENTS
Subject to the remaining budget, I would compare UCB with Expected Improvement on identical candidate pools and examine sensitivity to the exploration weight and kernel bounds. These are proposed comparisons, not completed results. I would record the configurations and costs and avoid treating an untested candidate's acquisition score as validation.

CONNECTION TO BROADER MACHINE LEARNING
This workflow resembles sequential experimental design in settings where evaluations are expensive, such as process tuning or laboratory optimisation. Its practical value depends on reproducible decisions, useful uncertainty estimates and careful separation of prediction from evidence. PCA and model comparisons support interpretation, but neither removes sampling bias nor proves that benchmark results transfer to a real industry.

COMMUNICATING TO A NON-TECHNICAL STAKEHOLDER
I would say: "We improved the best result in seven of eight tasks, but progress has slowed and the latest completed round produced no new best results. We choose each next test using both expected performance and what remains uncertain; the next eight outcomes are still unknown."

Evidence: CP_12-CP_23 notebooks and recorded outputs.
Project: github.com/BeatrizEchauz/bbo-capstone
The public repository may lag this local CP_23 snapshot.